This pdf file provides the supplementary material for our proposed federated deep learning framework for privacy-preserving clinical decision support. The framework integrates Vision Transformers (ViT) with self-supervised learning to enable robust medical image representation, particularly in scenarios with limited labeled data.

TABLE II: Nomenclature of Symbols Used in Methodology.

Symbol	Description
N	Number of clients in federated learning
R	Total number of communication rounds
θ^r	Global model at round r
θ_i^r	Local model of client i at round r
$\mathcal{D}_i$	Local dataset of client i
T	Number of local training epochs per round
b	Mini-batch size
d	Number of model parameters
z_i^s, z_i^t	Student and teacher ViT embeddings from DINO
$\mathcal{L}_{\text{DINO}}$	Self-supervised DINO loss function
μ	FedProx regularization coefficient
$\nabla \mathcal{L}_i^r$	Gradient of client i at round r
C	Gradient clipping threshold (for DP)
$\mathcal{N}(0, \sigma^2 I)$	Gaussian noise added for differential privacy
$\Delta\theta_i^r$	Model update of client i
$\hat{\Delta\theta}_i^r$	Encrypted update using ECC
$K_{\text{pub}}, K_{\text{priv}}$	Public and private keys used in ECC encryption
$\mathcal{O}(\cdot)$	Time or space complexity notation
p	Bit-size of the elliptic curve field in ECC
C (space complexity)	Number of output classes
$\mathcal{T}$	Data augmentation transformation set
μ, σ	Mean and standard deviation for normalization
x_i	Preprocessed image tensor
y_i	Ground truth label for image x_i
f	Original model (for LIME interpretability)
g	Interpretable surrogate model (LIME)
$L_{\text{Grad-CAM}}^c$	Grad-CAM saliency map for class c
α_k^c	Gradient-derived weights for channel k in class c
A^k	Activation map for channel k

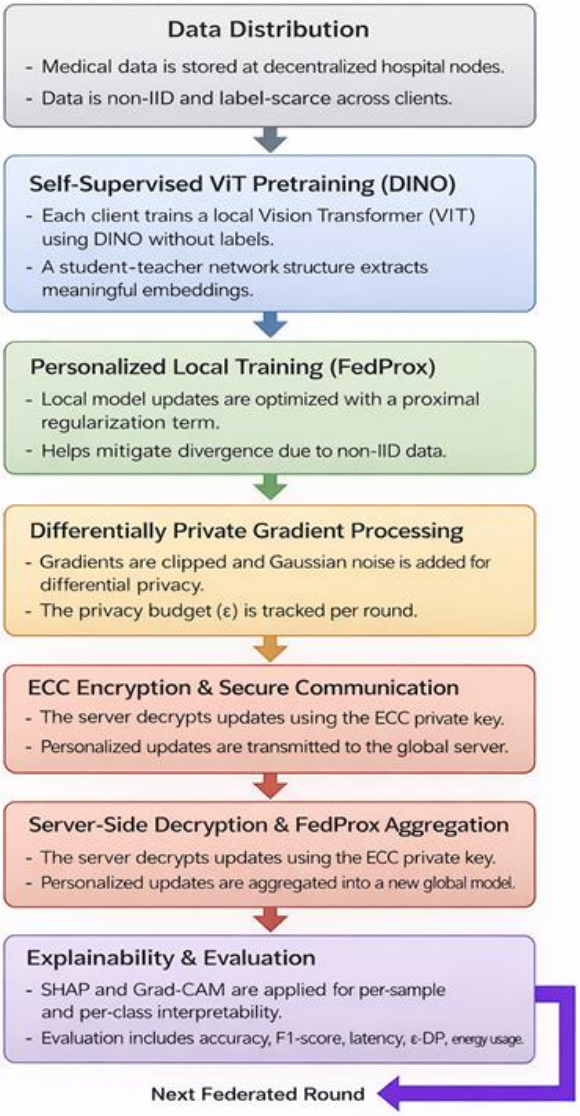

Fig. 2: Proposed FL Framework Flow for Privacy-Preserving AI-Driven Clinical Decision-Making.

TABLE III: Experimental Configuration and Reproducibility Parameters.

Parameter	Value
Hardware	Google Colab Pro (Tesla T4 GPU, 16 GB VRAM)
Framework	PyTorch 2.0.1
Python Version	3.10
Image Resolution	224×224
Preprocessing	CLAHE, rotation, horizontal flip, color jitter, normalization
Backbone Network	ViT-Base (patch size 16×16, input 224)
Self-Supervised Pretraining	DINOv2
Federated Rounds (R)	5
Number of Clients (N)	3 (TB, DR, BT)
Local Batch Size (TB)	16
Local Batch Size (DR)	8
Local Batch Size (BT)	12
Federated Aggregation	FedProx ($\mu = 0.01$)
Optimizer	Adam
Learning Rate	1×10^{-4} (federated), 1×10^{-5} (fine-tuning)
Loss Function	Cross-Entropy Loss
Differential Privacy Mechanism	DP-SGD (Opacus)
Noise Multiplier (σ)	1.3
Gradient Clipping Norm (C)	1.0
Epochs	100
Early Stopping Patience	5
Normalization	ImageNet mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]

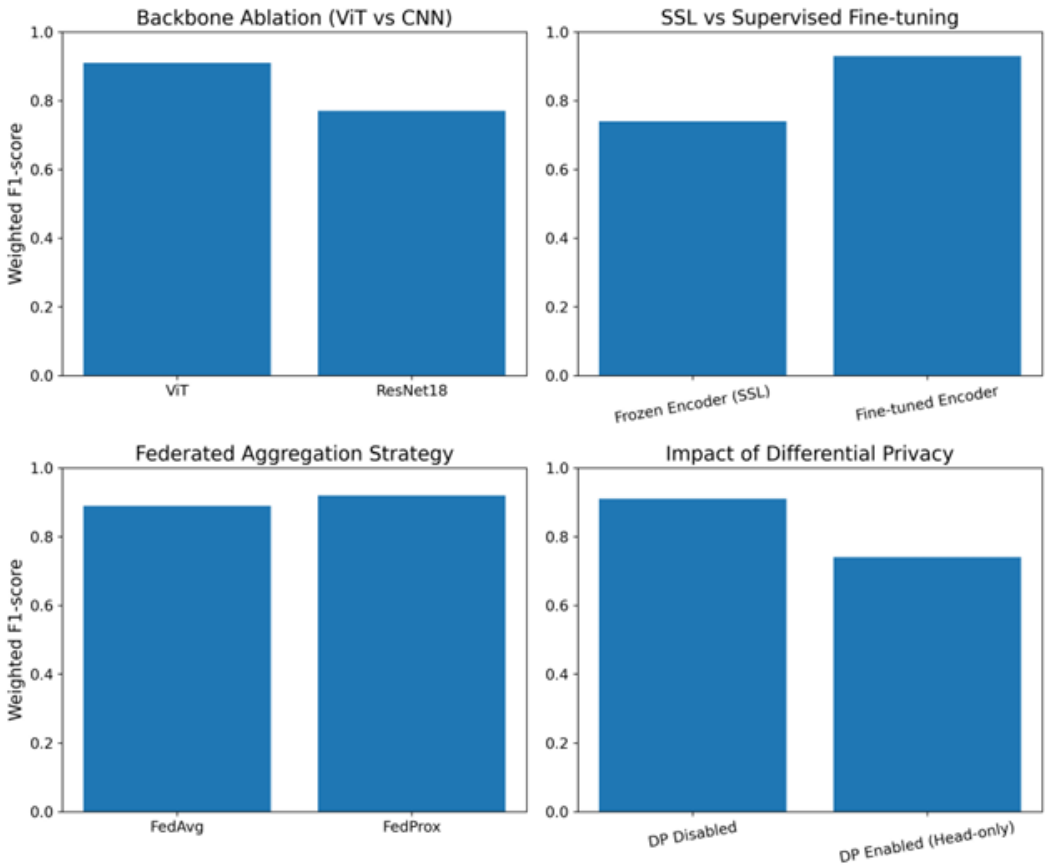

Fig. 15: Ablation study on key components of the proposed framework.